## 1. LINE NUMBERING SYSTEM

### EXAMPLE: 6"-P-1001-A1A

| | |
|---|---|
| **6"** | NOMINAL PIPE SIZE (INCH) |
| **P** | FLUID CODE |
| **1001** | SEQUENCE NO. (1st DIGIT = AREA) |
| **A1A** | PIPING CLASS |

## 2. FLUID CODES

| | |
|---|---|
| **P** | PROCESS |
| **CW** | COOLING WATER (SUPPLY/RETURN) |
| **LPS** | LOW PRESSURE STEAM |
| **N** | NITROGEN |
| **IA** | INSTRUMENT AIR |
| **D** | DRAIN |

## 3. PIPING CLASSES

| | |
|---|---|
| **A1A** | CS (A106 Gr.B), ASME 150#, CA 3.0 mm - PROCESS |
| **B1A** | CS GALVANIZED, ASME 150#, CA 1.5 mm - UTILITY WATER |
| **S1A** | SS316L, ASME 150#, CA 0 mm - CORROSIVE / PRODUCT |

## 4. EQUIPMENT IDENTIFICATION

| | |
|---|---|
| **TK** | TANK |
| **V** | VESSEL / DRUM / SEPARATOR |
| **P** | PUMP (A/B = OPERATING/STANDBY) |
| **E** | HEAT EXCHANGER / COOLER |
| **C** | COLUMN |
| **K** | COMPRESSOR |

## 5. INSTRUMENT IDENTIFICATION LETTERS (ISA 5.1)

| 1ST LETTER (MEASURED VARIABLE) | | SUCCEEDING LETTERS (FUNCTION) | |
|---|---|---|---|
| **F** | FLOW | **I** | INDICATOR |
| **L** | LEVEL | **C** | CONTROLLER |
| **P** | PRESSURE | **T** | TRANSMITTER |
| **T** | TEMPERATURE | **V** | VALVE |
| | | **SV** | SAFETY VALVE |

## 6. SYMBOLS


PROCESS LINE


UTILITY LINE

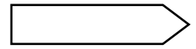

INSTRUMENT SIGNAL

CONTROL VALVE

SAFETY / RELIEF VALVE

FIELD INSTRUMENT

PANEL INSTRUMENT (DCS)


OFF-PAGE CONNECTOR

## 7. GENERAL NOTES

1. ALL DIMENSIONS IN MM UNLESS OTHERWISE NOTED.
2. LINES WITHOUT INSULATION CODE ARE NOT INSULATED.
3. STANDBY PUMPS SHARE THE EQUIPMENT DATA OF THE OPERATING PUMP.

**P&ID LEGEND AND SYMBOLS**

PIPING AND INSTRUMENT DIAGRAM - SAMPLE PROJECT

| DWG NO. | REV. |
|---|---|
| **PID-100-000** | **0** |